Machine Learning

08-04-26

Multi-linear Regression

hypothesis function: $y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$

loss function: $L = MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$$L = \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + b) \right]^2$$

find optimal values of w_1 w_2 w_3 and b using gradient descent.

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

$$L = \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + b) \right]^2$$

put $z_i = y_i - (w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + b)$

$$L = \frac{1}{n} \sum_{i=1}^{n} (z_i)^2$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_i} \times \frac{\partial z_i}{\partial w_1} \quad \cdots \text{ by chain rule}$$

$$= \frac{\partial \left(\frac{1}{n} \sum_{i=1}^{n} z_i^2 \right)}{\partial z_i} \times \frac{\partial \left[y_i - (w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + b) \right]}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial z_i^2}{\partial z_i} \times \frac{\partial (y_i - w_1 x_{i1} - w_2 x_{i2} - w_3 x_{i3} - b)}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial z_i^2}{\partial z_i} \times \frac{\partial (y_i - w_1 x_{i1} - w_2 x_{i2} - w_3 x_{i3} - b)}{\partial w_1}$$

$$= \frac{1}{n}\sum_{i=1}^{n} 2z_i \times \left[\frac{\partial y_i}{\partial w_1} - \frac{\partial w_1 x_{i1}}{\partial w_1} - \frac{\partial w_2 x_{i2}}{\partial w_1} - \frac{\partial w_3 x_{i3}}{\partial w_1} - \frac{\partial b}{\partial w_1}\right]$$

$$= \frac{1}{n}\sum_{i=1}^{n} 2z_i \times \left[0 - x_{i1} - 0 - 0 - 0\right]$$

$$\frac{\partial L}{\partial w_1} = \frac{2}{n}\sum_{i=1}^{n} z_i(-x_{i1})$$

$$\boxed{\frac{\partial L}{\partial w_1} = \frac{-2}{n}\sum_{i=1}^{n} z_i x_{i1}}$$

$$\frac{\partial L}{\partial w_2} = \frac{-2}{n}\sum_{i=1}^{n} z_i x_{i2}$$

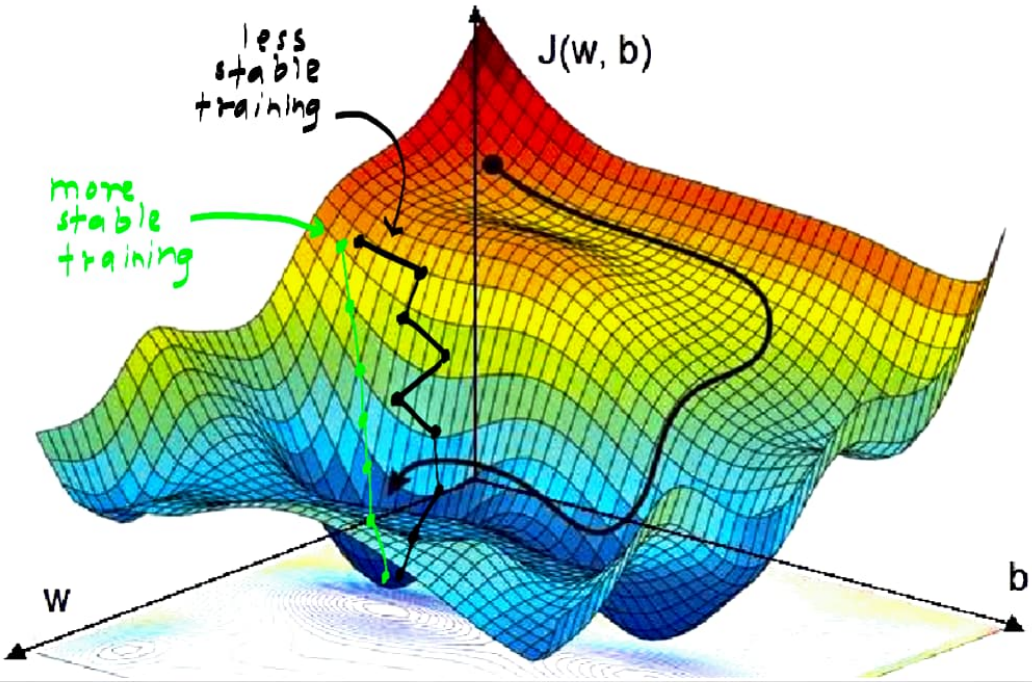
$$\frac{\partial L}{\partial w_3} = \frac{-2}{n}\sum_{i=1}^{n} z_i x_{i3}$$

$$\frac{\partial L}{\partial b} = \frac{-2}{n}\sum_{i=1}^{n} z_i = \frac{\partial L}{\partial z_i} \times \frac{\partial z_i}{\partial b} = \frac{2}{n}\sum_{i=1}^{n} z_i \times (-1) = -\frac{2}{n}\sum_{i=1}^{n} z_i$$

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

$$b_{new} = b_{old} - \alpha \frac{\partial L}{\partial b_{old}}$$

keep updating model parameters until convergence



- In stable training, model takes shortest and steepest steps towards minima.
- Due to this, training becomes smoother and faster

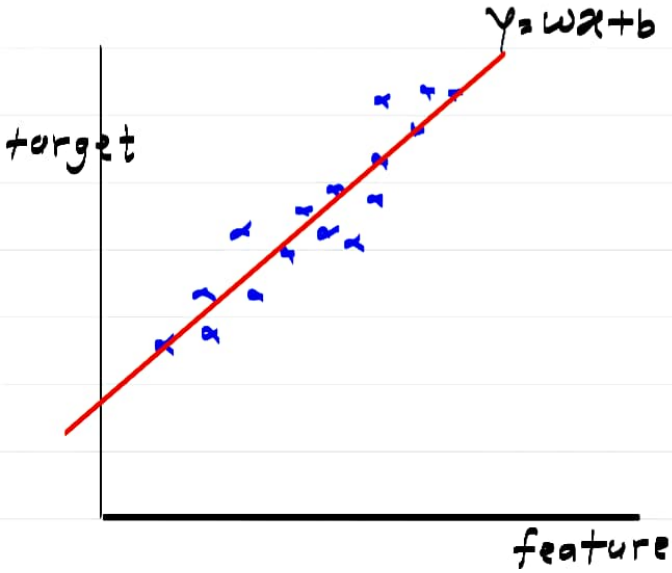
$$\frac{\partial L}{\partial w_1} = \frac{-2}{n} \sum_{i=1}^{n} z_i x_{i1} \qquad x_{i1} \longrightarrow \text{range } 0-1$$

$$\frac{\partial L}{\partial w_2} = \frac{-2}{n} \sum_{i=1}^{n} z_i x_{i2} \qquad x_{i2} \longrightarrow \text{range } 100-1000$$

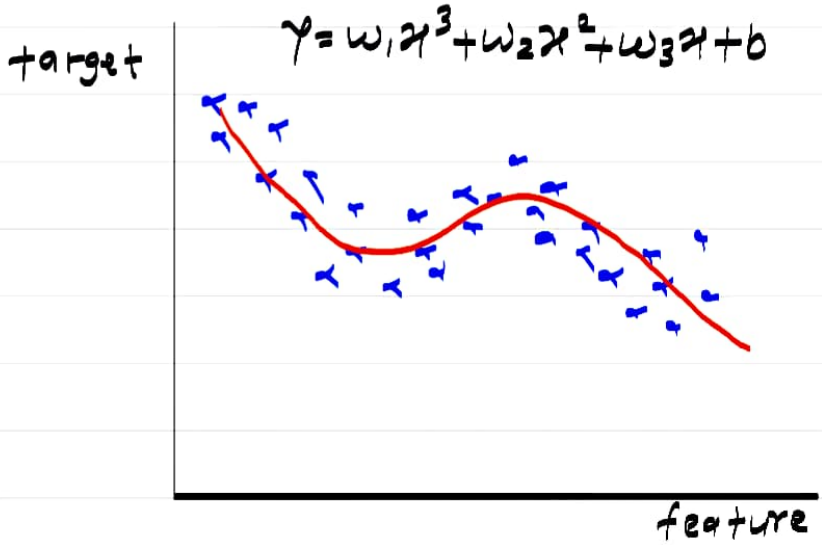
$$\left| \frac{\partial L}{\partial w_2} \right| >> \left| \frac{\partial L}{\partial w_1} \right|$$

- due to this w_2 will get larger updates than w_1 and training gets biased towards w_2
- To avoid biased weight updation and unstable training, feature scaling is must before gradient descent.

Polynomial Regression



linear regression



polynomial regression

To train a polynomial regression model, we need to do feature engineering and feature transformation

$$\underset{\text{feature}}{\underline{x}} \quad \underset{\text{target}}{\underline{y}} \xrightarrow{\text{polynomial degree} = 3} \underbrace{x \quad x^2 \quad x^3}_{\text{features}} \quad \underset{\text{target}}{\underline{y}}$$

$y = wx + b$

$$y = w_1 x + w_2 x^2 + w_3 x^3 + b$$

$$\downarrow$$

$$y = w_1 f_1 + w_2 f_2 + w_3 f_3 + b$$

$$\downarrow$$

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

hyp. func: $y = w_1 f_1 + w_2 f_2 + w_3 f_3 + b$

$$\frac{\partial L}{\partial w_1} = -\frac{2}{n} \sum_{i=1}^{n} z_i f_{i1} \qquad \frac{\partial L}{\partial w_2} = -\frac{2}{n} \sum_{i=1}^{n} z_i f_{i2}$$